

Shri Shamrao Patil (Yadravkar) Educational & Charitable Trust's



Sharad Institute of Technology

College of Engineering, Yadrav (Ichalkaranji)

An Autonomous Institute

Department of Electronics and Computer Engineering

A
Mini Project Synopsis
on
SMART SHOPPING TROLLEY

SUBMITTED BY

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(Approved by AICTE, New Delhi, Recognized by Government of Maharashtra & Affiliated to Dr. Babasaheb
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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

This is to certify that,

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Studying in TY ECE has successfully completed the Mini Project Synopsis **“SMART SHOPPING TROLLEY”** under the guidance and supervision of Ms.M.B.Mulik during the academic year 2025-2026 Semester V. This is a part of partial fulfillment of the requirement for award of degree of the Bachelor of Technology in Electronics and Computer Engineering.

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ABSTRACT

Conventional billing systems used in the supermarket nowadays are a big hassle for the customers as it creates a long waiting time for payment of the products purchased. To provide a solution for the above problem, we have designed an application called Smart Shopping Cart using IoT which can eliminate queues for billing throughout the supermarket. Using the application, we scan the products and then put them inside a Smart cart which is designed using Node MCU, and LCD to display the total price of the items in the cart. To make it more feasible the bill is also made available on the web server that makes it easy for the customer to check the list of items added and so they can manage the purchase. The system enables real-time tracking of each item added to the cart, giving shoppers immediate visibility into their spending and allowing them to stay within budget. It also streamlines the retailer's workflow by reducing reliance on manual checkout counters, thereby improving throughput and customer satisfaction. With the integration of a web interface, shoppers can review their cart content remotely, receive digital receipts, and even share the bill with others if needed. The IoT-enabled cart architecture is modular and scalable, so it can be deployed across different store sections with minimal reconfiguration. Finally, this approach opens the door for future enhancements such as dynamic offers, targeted coupons and inventory integration, making the supermarket shopping experience smarter and more efficient.

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INTRODUCTION

In our day-to-day life, we see that the use of computing technology and innovative applications have increased rapidly. Many people are connected to this digital world through their smartphones and laptops because of the high speed internet connectivity, user-friendly interface, and personalized recommendation. This has made things simpler and easier. The new technology such as social media has given the users a better and broader perspective of many items and information about various products in their daily life and ensures that they are of the best quality. As our cities are overpopulated, the diversity of wants and needs of the people are also huge. As a result, many people prefer to go to malls and supermarkets to buy different varieties of commodities. The major drawback here is the people coming to shopping malls and supermarkets are huge in number, resulting in large queues for billing. Customers have to wait for a long time in a queue for scanning their items and paying the bill. An automated smart shopping system is formed by introducing the concept of IoT to connect all items in the grocery shop. In this system, an inexpensive RF-ID tag is embedded within each product. When the product is placed into a smart cart, the product detail is automatically read by the cart equipped with an RF-ID reader. Hence, billing is made from the shopping cart itself preventing customers from waiting in a long queue at checkout. If any item is not scanned due to tampered bar code, we can identify that based on weight at last. Thus, inventory management becomes easier. This system makes both the customer and supermarket employee's work easier by reducing the time it actually takes while using the old method of billing i.e manual billing technique.

LITERATURE SURVEY

[1] Kumar, A. Gupta, S. Balamurugan, S. Balaji and R. Marimuthu, "Smart Shopping Trolley," 2017 International conference on Microelectronic Devices, Circuits and Systems (ICMDCS), 2017, pp. 1-4. This paper proposes a system that reduces queues at billing counters in shopping complexes by displaying the total price of items in the trolley. The system eliminates traditional scanning at the counter, speeds up shopping, and reduces the possibility of human error. It is based on the Arduino platform and Xbee modules for wireless automation.

[2] V. V., P. K. P. and C. R. S., "Smart Shopping Trolley," 2018 International Conference on Circuits and Systems in Digital Enterprise Technology (ICCSDET), 2018, pp. 1-4. Our proposed solution to reduce shopping time and improve customer experience at supermarkets involves a smart shopping trolley equipped with a barcode scanner and touchscreen display. This allows customers to scan products and view information and cost in realtime, culminating in a total bill displayed on the trolley. Payment can then be made through various online options such as Paytm, UPI, or PhonePay.

[3] T. K. Das, A. K. Tripathy and K. Srinivasan, "A Smart Trolley for Smart Shopping," 2020 International Conference on System, Computation, Automation and Networking (ICSCAN), 2020, pp. 1-5. Our proposed solution is an IoT-based system consisting of a smart trolley equipped with an Arduino board, RFID reader, RFID tag, LCD display, ESP8266 Wi-Fi module, and a website to maintain product and customer details. The trolley can interact with a network spread worldwide, allowing customers to purchase products, receive e-bills through email, and view purchase details through the shop's website.

[4] M. Jaishree., K. R. Lakshmi prabha., S. Jeyaprabha. and K. Mohan., "Smart Shopping Trolley Using IOT," 2021 7th International Conference on Advanced Computing and Communication Systems. This paper proposes an automated shopping trolley with a barcode scanner and LCD display, using the Raspberry Pi device. It aims to reduce waiting time in queues during rush hours at supermarkets.

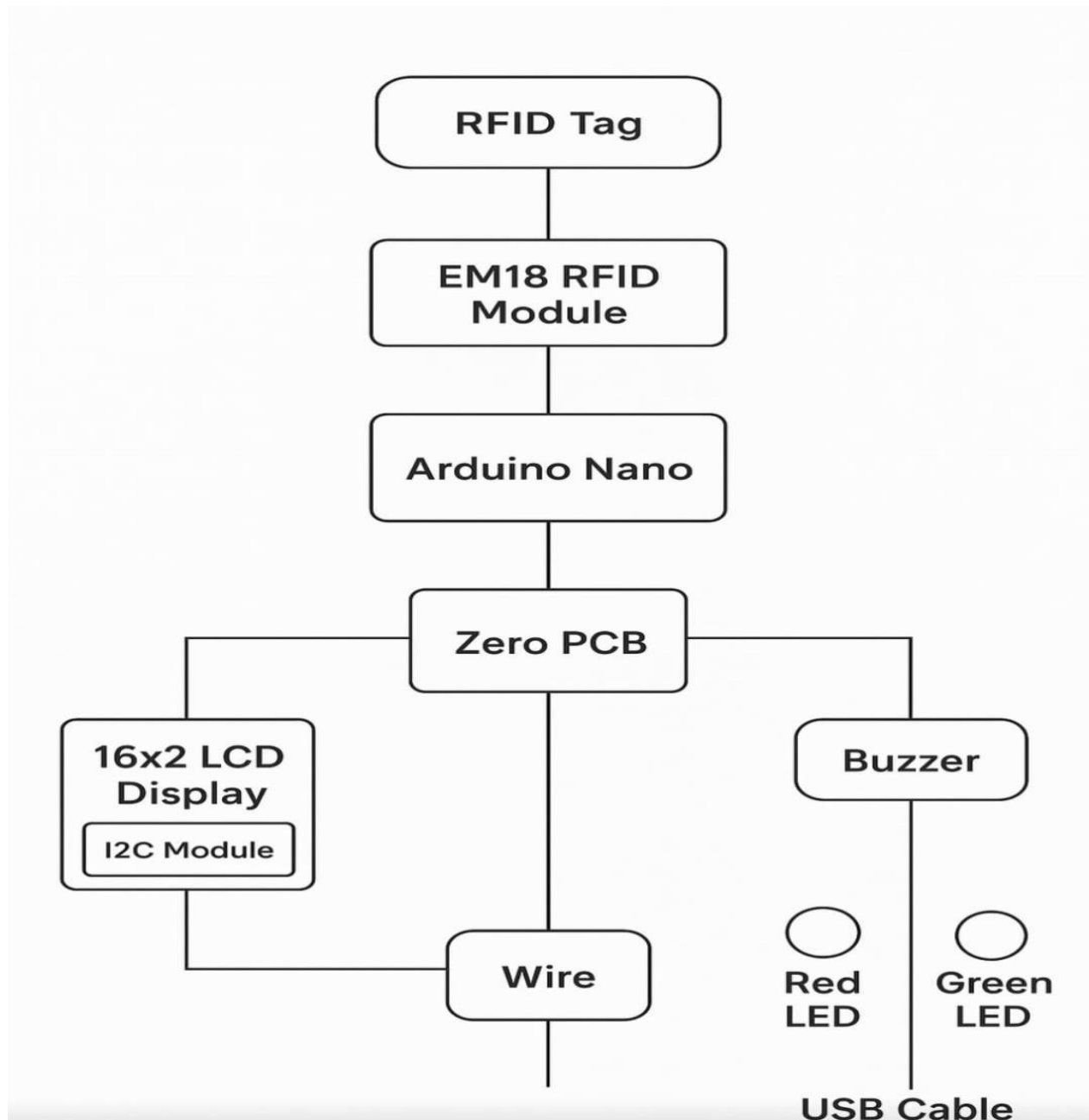
[5] M. Shahroz, M. F. Mushtaq, M. Ahmad, S. Ullah, A. Mehmood and G. S. Choi, "Smart Shopping Trolley Using Radio Frequency Identification," in IEEE Access, vol. 8, pp. 68426-68438. The proposed IoT-based Smart Shopping Trolley utilizes RFID sensors, Arduino microcontroller, Bluetooth module and Mobile application to efficiently manage Shopping.

OBJECTIVE

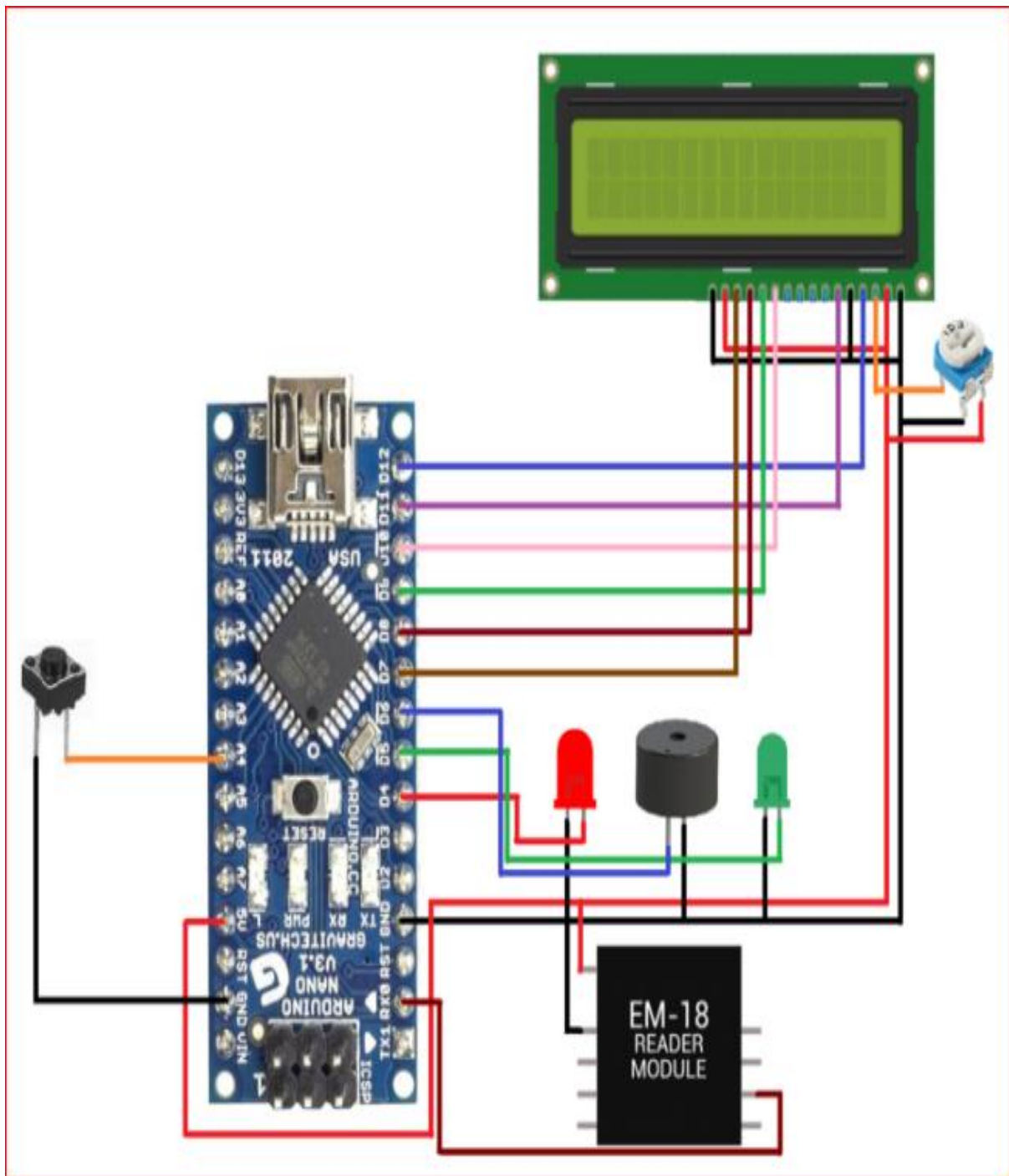
The objectives are detailed as follows,

1. To find an efficient way to implement this cart feasible.
2. To scan and update the price of item in webserver.
3. To have a detailed bill description.

BLOCK DIAGRAM



CIRCUIT DIAGRAM



COMPONENT DESCRIPTION

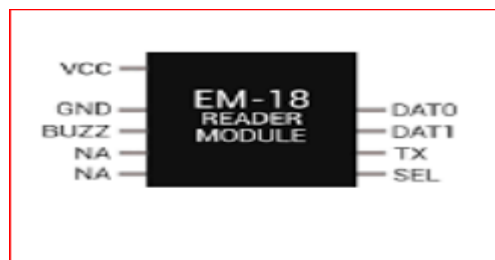
1. Arduino Nano Board

The Arduino Nano is a compact, breadboard-friendly microcontroller board based on the ATmega328P chip (in many cases) and developed by Arduino.cc. It operates typically at 5 V logic (with a 16 MHz clock) and offers many digital and analog I/O pins in a small form factor. The board supports USB connection for programming and power, and can also be powered externally via VIN. Because of its versatility and size, it is ideal for embedded & prototype projects like your smart shopping trolley. It handles reading inputs (such as from sensors or modules), processing logic (e.g., item scanning, totaling), and controlling outputs (LCD, LEDs, etc.). It integrates with the Arduino IDE which simplifies firmware development if your project is programmed in Arduino style. In summary, the Arduino Nano serves as the “brain” of your system, coordinating all other components.



2. EM18 RFID Module

The EM18 RFID reader module is a low-cost, widely used reader operating at 125 kHz frequency and designed to detect compatible RFID tags within a short range (~5–10 cm) depending on environment and tag. It outputs the tag's unique ID via a serial TTL (UART) interface making it easy to connect with microcontrollers. The module typically works with a 5 V supply and consumes around <50 mA current. Its compact size and simple interface make it convenient for embedding into your smart cart for item-tag scanning. It acts as the interface between the physical RFID tags on products and the microcontroller logic. When a tag is brought into its antenna field, it reads the ID and relays that to the Arduino Nano to lookup product info and update the total.



3. RFID Tag

RFID tags are small devices attached to each product in your smart shopping cart system. When a tag comes within range of the EM18 RFID reader module's antenna, the reader powers the tag and receives its unique identification number. The reader then sends that ID to the microcontroller for processing. In your application, each tag uniquely identifies an item/product, enabling the microcontroller to determine price, product name, etc., and update the display and bill accordingly. These tags require no onboard power and are inexpensive, making them practical for bulk use in a supermarket environment. By using RFID tags, your system can automatically detect items placed in the cart without manual barcode scanning, thereby reducing billing queue time.



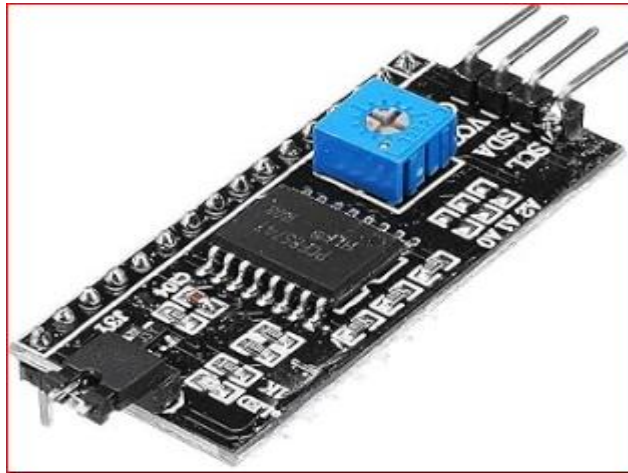
4. 16×2 LCD Display (with I2C module)

This is a 2-line by 16-characters (16×2) alphanumeric liquid-crystal display module, used to show information such as item list, running total cost, or instructions to the shopper. When combined with an I2C interface module, the wiring is simplified dramatically of many pins, only four are needed: VCC, GND, SDA, SCL. The I2C module uses an I/O expander to translate I2C bus signals into parallel signals that drive the LCD. This arrangement frees up many MCU pins and simplifies code and wiring. In your smart cart, the LCD provides immediate visual feedback to the customer: showing scanned items, displaying the total, alerting if there's an error, etc. It's especially helpful for user-interaction in real time inside the cart. You will need to include contrast/backlight adjustment so that display readability is good under store lighting conditions.



5. I2C Module (for LCD)

The I2C module is a small adapter board mounted on the back of the 16×2 LCD which converts the multiple parallel interface lines of a standard LCD into an I2C serial interface. The module typically uses the PCF8574 or similar I/O-expander chip and allows control of the LCD via the SDA and SCL lines of the I²C bus. It reduces the number of required microcontroller pins significantly for controlling the display module. In your project, that leaves many I/O pins free on the Arduino Nano for other controls. The module also usually provides solder-jumpers or address lines so multiple I2C devices can share the same bus. It simplifies wiring, especially in a compact design like a smart shopping cart. By using I2C connectivity, the system becomes more modular and easier to expand in future iterations.



6. Buzzer

A buzzer is an audio signalling device which, when triggered by the microcontroller output, emits a sound (buzz, beep, alarm) to alert or inform the user. It can be piezoelectric or electromagnetic in design. In your smart shopping cart system, the buzzer serves as a feedback mechanism — for example, when an item is scanned/plucked, when an error occurs (e.g., invalid tag), or when checkout is complete. It draws very little current and is simple to interface via a microcontroller digital output pin (optionally via a transistor). Including auditory feedback enhances the user experience in a busy store environment where visual cues alone may not suffice. You can program different beep patterns for different events (e.g., short beep for item added, longer beep for error) for clarity.



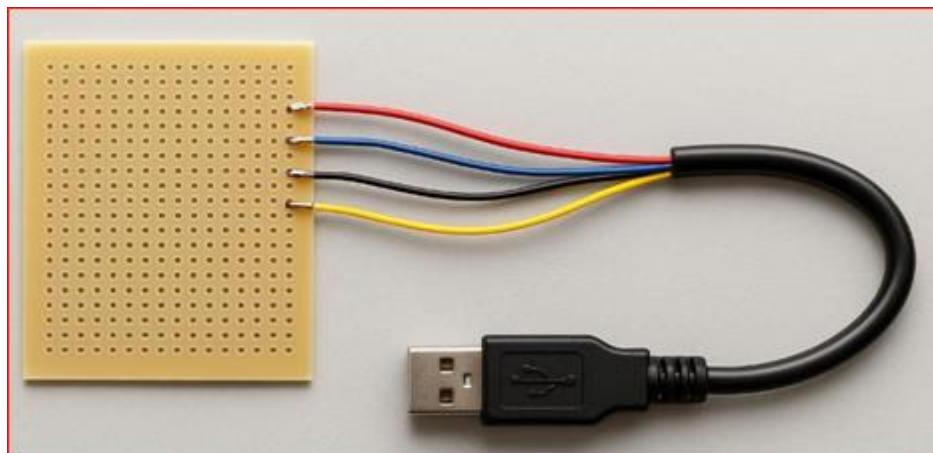
7. Red LED & Green LED

LEDs (Light Emitting Diodes) are simple semiconductor devices that emit light when current passes through them in the forward direction. In your design, the green LED can indicate *successful item scan or valid tag added*, while the red LED can indicate *error, tag not recognised, or unauthorized item*. These visual indicators provide instant, easy-to-understand status feedback for both the user and the system. Because the LEDs draw very little current, they are ideal for being lit intermittently without significant impact on power consumption. By using colour codes (green for “OK”, red for “Attention/Error”), the interface becomes intuitive and reduces dependence on text or explanation.



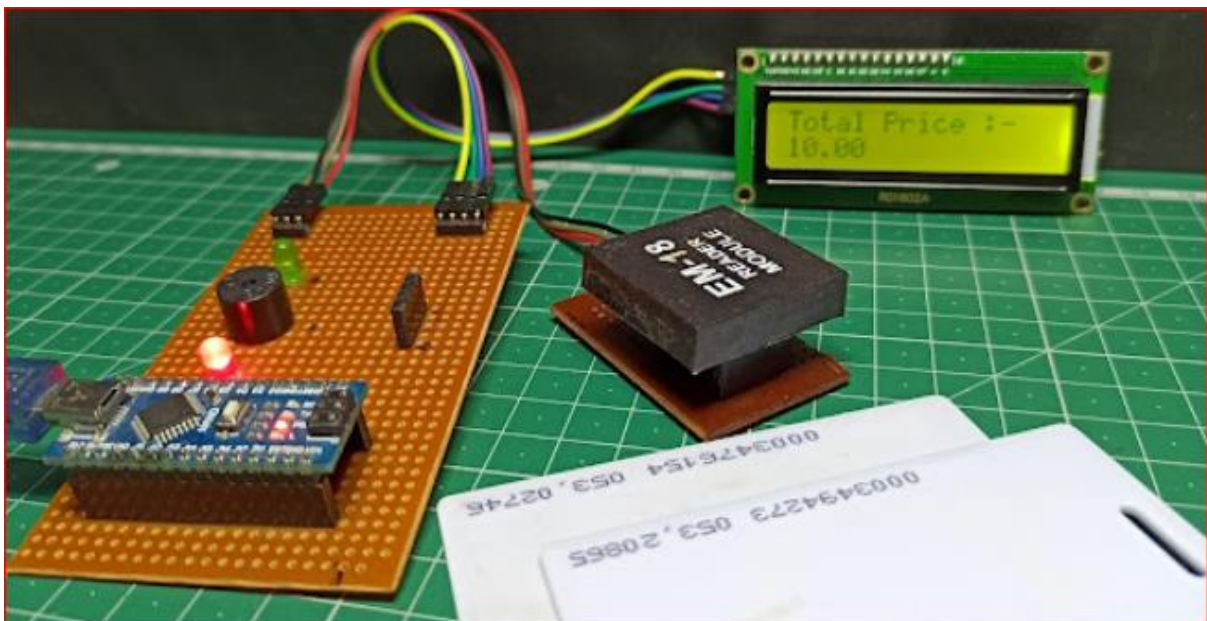
8. Zero PCB (Prototype Board) + Wires + USB Cable

The zero PCB (also called prototyping board or breadboard-friendly PCB) provides a means to physically mount and interconnect the various components: the Arduino Nano, RFID module, LCD & I2C module, buzzer, LEDs, power wiring, etc. Using a PCB or well-organized wiring layout improves reliability, durability and neatness of the prototype/cart build. The wires serve to connect components: power lines (5 V, GND), signal lines . USB cable is used to both supply power (if you use a USB power bank or store supply) and to program the Arduino Nano. In your smart cart system, the USB cable may also allow data uplink or simply power the board from the store’s USB power source. Good wiring practices help avoid faults such as loose connections, interference or voltage drops, which is especially important in a mobile cart environment. Also ensure power supply is stable and sufficient to drive all components without brown-outs.



CIRCUIT DESIGN

In the circuit design of the smart shopping trolley system, the core control element is the Arduino Nano microcontroller which interfaces with the scanning, display and feedback modules. The EM18 RFID Module is connected to the Nano to detect each item's RFID tag as it enters the cart; its serial output is fed into one of the Arduino's digital pins so that each tag's unique ID can be processed. Meanwhile, the 16×2 LCD display (via an I2C interface) is connected over the Arduino's I²C bus (SDA, SCL) to display the list of scanned items and the running total cost, minimizing pin usage and simplifying wiring. Visual cues are provided by the red and green LEDs wired to digital output pins (with appropriate current-limiting resistors) to indicate status (error / success), and an audio buzzer is driven by the Arduino to alert the shopper when needed. All modules share a common 5 V power rail and GND line (ensuring proper reference), and the wiring is mounted on a zero-PCB for neatness and reliability in a mobile cart environment. Care has also been taken to ensure that signal and power lines are appropriately routed and kept short to reduce noise or voltage drop issues, which is especially important in a real-time store environment. This design allows the microcontroller to carry out the core logic — reading the tag, looking up its price, updating the total, updating the display, activating feedback — all with clean wiring and modular connections, thereby enabling robust and scalable implementation of the smart trolley concept.



ADVANTAGES

Smart shopping carts offer several advantages over traditional shopping carts. Here are some of the key advantages:

- **Improved Shopping Experience:** Smart shopping carts can enhance the shopping experience by providing features such as product recommendations, real-time promotions, and personalized coupons based on the customer's shopping history.
- **Time Savings:** Smart shopping carts can save time for both customers and retailers. Customers can quickly and easily find the products they need, while retailers can streamline checkout processes and reduce the time spent on manual inventory tracking.
- **Increased Efficiency:** Smart shopping carts can help retailers increase efficiency by automating tasks such as inventory management, pricing, and restocking. This can free up employees to focus on other tasks, such as customer service.
- **Enhanced Security:** Smart shopping carts can improve security by incorporating features such as RFID tags and weight sensors to prevent theft or unauthorized removal of products from the store.
- **Data Insights:** Smart shopping carts can provide retailers with valuable data insights, such as customer behavior, preferences, and trends. This can help retailers make informed decisions about pricing, promotions, and product placement.
- **Reduced Costs:** Smart shopping carts can help reduce costs for retailers by eliminating the need for manual inventory tracking, reducing waste through real-time inventory management, and reducing the time and cost associated with checkout processes.
- **Overall,** smart shopping carts can provide a more efficient and personalized shopping experience for customers, while also providing valuable data insights and cost savings for retailers.

APPLICATIONS

Smart shopping carts have several applications in retail and other industries. Here are some of the most common applications:

- Retail: Smart shopping carts can be used in retail stores to enhance the shopping experience for customers. They can provide real-time product recommendations, personalized coupons and promotions, and streamlined checkout processes.
- Supermarkets: Smart shopping carts can be used in supermarkets to improve inventory management and reduce waste. They can track inventory levels in real-time and alert staff when products need to be restocked.
- Hospitality: Smart shopping carts can be used in hotels and resorts to provide guests with a convenient way to order room service and other amenities. Guests can use the smart cart to browse menus, place orders, and track their purchases.
- Airports: Smart shopping carts can be used in airports to provide travelers with a convenient way to shop for duty-free items and other products. They can be equipped with RFID tags to track inventory levels and prevent theft.
- Smart Cities: Smart shopping carts can be used in smart cities to track waste disposal and recycling. They can be equipped with sensors to monitor waste levels and alert staff when bins need to be emptied.
- Healthcare: Smart shopping carts can be used in healthcare settings to improve medication management. They can be equipped with RFID tags to track medication inventory levels and alert staff when medications need to be restocked.
- Overall, smart shopping carts have numerous applications in various industries, providing businesses with a way to improve efficiency, enhance the shopping experience for customers, and reduce waste and costs.

CONCLUSION

In conclusion, smart shopping carts have the potential to revolutionize the shopping experience by providing real-time information and personalized recommendations to customers while also providing valuable data to retailers. The basic idea is to make shopping more easy and comfortable in the overpopulated cities. This shopping cart is built with a system that enables customer to bill their products at cart itself without standing in long ques for billing. Finally, a system named smart shopping cart is designed in which the microcontroller(node mcu) and rfid technology plays a vital role in scanning the products,preparing bill and displaying it on lcd and also uploading the same into the webserver .This helps the user to view the bill on his/her device.

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